School of Information Technology and Engineering

Security of Operating System

SIS 5

Monitoring & Observability

“Secure SWIFT Logical Terminal Infrastructure”

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Topic

Monitoring & Observability

Target

The goal of this study is to design and implement a monitoring, logging, and auditing system for the Secure SWIFT Logical Terminal Infrastructure.

This includes:

- Centralized logging system
- Log analysis tooling (scripts)
- Security auditing using Linux audit subsystem

This study continues SIS1–SIS4 and focuses on observability, traceability, and security monitoring.

1. Defining Logging Mechanism

1.1 Logging Architecture Overview

The system uses centralized logging:

```
swift-terminal      ───────▶      swift-support
   (client)                        (log server)
```

Flow:

- All logs are generated on swift-terminal
- Logs are forwarded using rsyslog
- Logs are stored centrally on swift-support

1.2 Log Sources (swift-terminal)

Component	Log Location
System logs	journald

SSH/Auth	/var/log/secure
Nginx	/var/log/nginx/
Backend (FastAPI)	journalctl (systemd)
PostgreSQL	/var/lib/pgsql/data/log
Firewall	firewalld logs

1.3 Central Log Storage (swift-support)

All logs are stored in:

Dir = /var/log/remote/

1.4 Logging Design Principles

- Centralization (logs not stored locally only)
- Separation (logs stored on different machine)
- Integrity (logs cannot be modified by attacker on main server)
- Auditability (all actions traceable)
- Minimal exposure (only rsyslog port open)

2. Setting up Logging and Tooling

2.1 Configure rsyslog (swift-support — SERVER)

Install:

sudo dnf install rsyslog -y

```
^Cswift-support@fedora:/$ sudo dnf install rsyslog -y
Updating and loading repositories:
 grafana                       100% |   1.6 KiB/s |   3.0 KiB |  00m02s
 Fedora 43 - x86_64 - Update 100% |   1.2 KiB/s |  11.5 KiB |  00m09s
^Cswift-support@fedora:/sudo dnf install rsyslog -y
Updating and loading repositories:
 grafana                       100% |   2.5 KiB/s |   3.0 KiB |  00m01s
 Fedora 43 - x86_64 - Update 100% |   3.9 KiB/s |  11.5 KiB |  00m03s
Repositories loaded.
Package "rsyslog-8.2508.0-1.fc43.x86_64" is already installed.

Nothing to do.
swift-support@fedora:/$
```

Edit config:

sudo nano /etc/rsyslog.conf

```
# Use default timestamp format
module(load="imudp")
input(type="imudp" port="514")

module(load="imtcp")
input(type="imtcp" port="514")
```

Create log directory:

sudo mkdir -p /var/log/remote

```
swift-support@fedora:/$ sudo mkdir -p /var/log/remote
```

Restart:

sudo systemctl restart rsyslog

```
swift-support@fedora:/$ sudo systemctl restart rsyslog
swift-support@fedora:/$
```

2.2 Configure Firewall (swift-support)

sudo firewall-cmd --permanent --add-port=514/udp
sudo firewall-cmd --permanent --add-port=514/tcp
sudo firewall-cmd –reload

```
swift-support@fedora:/$ sudo firewall-cmd --permanent --add-port=514/
udp
success
swift-support@fedora:/$ sudo firewall-cmd --permanent --add-port=514/
tcp
success
swift-support@fedora:/$ sudo firewall-cmd --reload
success
swift-support@fedora:/$
```

2.3 Configure rsyslog (swift-terminal — CLIENT)

Edit:

sudo nano /etc/rsyslog.conf

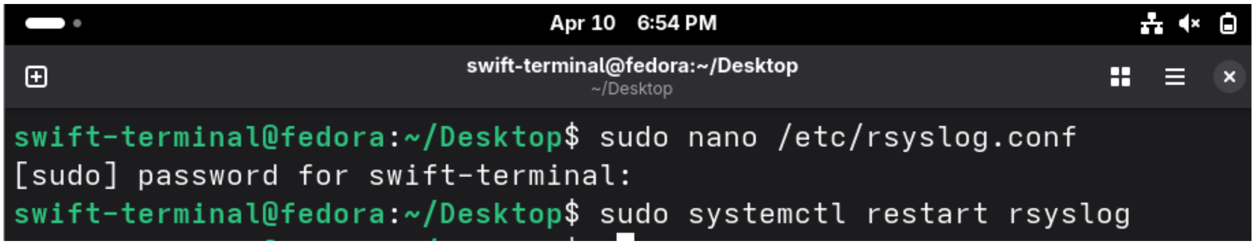
Add:

. @<swift-support-IP>:514

```
# ### sample forwarding rule ###
#action(type="omfwd"
# # An on-disk queue is created for this action. If the remote host >
# # down, messages are spooled to disk and sent when it is up again.
#queue.filename="fwdRule1"       # unique name prefix for spool files
#queue.maxdiskspace="1g"         # 1gb space limit (use as much as p>
#queue.saveonshutdown="on"       # save messages to disk on shutdown
#queue.type="LinkedList"         # run asynchronously
#action.resumeRetryCount="-1"    # infinite retries if host is down
# # Remote Logging (we use TCP for reliable delivery)
# # remote_host is: name/ip, e.g. 192.168.0.1, port optional e.g. 10>
#Target="remote_host" Port="XXX" Protocol="tcp")
*.* @10.156.104.194:514
```

Restart:

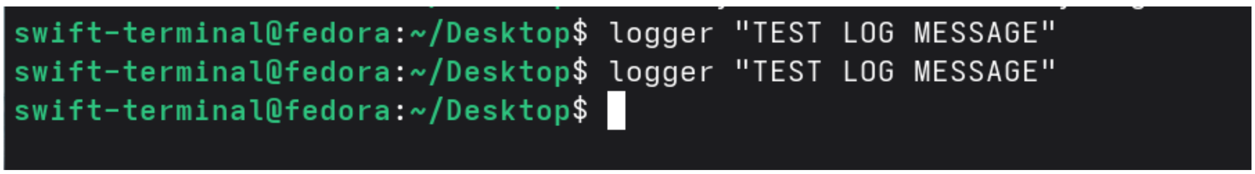
sudo systemctl restart rsyslog



2.4 Verification

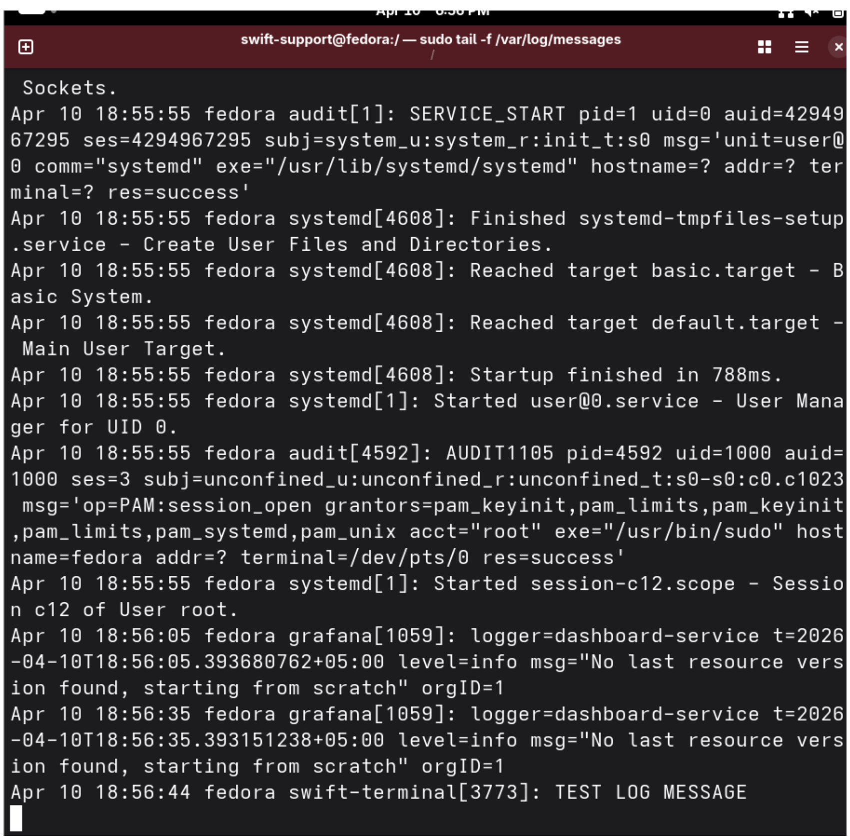
Generate test log:

logger "TEST LOG MESSAGE"



Check on support server:

sudo tail -f /var/log/messages



2.5 Log Analysis Scripts (Tooling)

```
swift-support@fedora:/home$ sudo mkdir -p /opt/scripts
swift-support@fedora:/home$ cd ./opt/scripts
```

Script 1 — Failed Login Detection

#!/bin/bash
echo "Failed SSH Logins:"
grep "Failed password" /var/log/secure | tail -n 20

```
  GNU nano 8.5                    fialed-login.sh
#!/bin/bash
echo "Failed SSH Logins:"
grep "Failed password" /var/log/secure | tail -n 20
```

Script 2 — Nginx Error Monitor

#!/bin/bash
echo "Nginx Errors:"
tail -n 20 /var/log/nginx/error.log

```
  GNU nano 8.5                    nginx-error.sh
#!/bin/bash
echo "Nginx Errors:"
tail -n 20 /var/log/nginx/error.log
```

Script 3 — SWIFT File Access Monitor

#!/bin/bash
echo "Access to /swift-lt:"
journalctl | grep "/swift-lt" | tail -n 20

```
  GNU nano 8.5                swift-file-access.sh
#!/bin/bash
echo "Access to /swift-lt:"
journalctl | grep "/swift-lt" | tail -n 20
```

Make scripts executable

chmod +x script.sh

```
swift-support@fedora:/opt/scripts$ sudo nano fialed-login.sh
swift-support@fedora:/opt/scripts$ sudo nano nginx-error.sh
swift-support@fedora:/opt/scripts$ sudo nano swift-file-access.sh
swift-support@fedora:/opt/scripts$ sudo chmod +x fialed-login.sh
swift-support@fedora:/opt/scripts$ sudo chmod +x nginx-error.sh
swift-support@fedora:/opt/scripts$ sudo chmod +x swift-file-access.sh
```

Run:

```
swift-support@fedora:/opt/scripts$ ./fialed-login.sh
Failed SSH Logins:
grep: /var/log/secure: Permission denied
swift-support@fedora:/opt/scripts$ sudo ./fialed-login.sh
Failed SSH Logins:
swift-support@fedora:/opt/scripts$ sudo ngettext
ngettext: missing arguments
swift-support@fedora:/opt/scripts$ sudo ./nginx-error.sh
Nginx Errors:
tail: cannot open '/var/log/nginx/error.log' for reading: No such fil
e or directory
swift-support@fedora:/opt/scripts$ sudo ./swift-file-access.sh
Access to /swift-lt:
Mar 27 23:26:23 fedora sudo[11370]:  backup1 : user NOT in sudoers ;
TTY=pts/1 ; PWD=/home/backup1 ; USER=root ; COMMAND=/usr/sbin/tar -xz
f /backup/swift/swift-lt-2026-03-27.tar.gz -C /
Mar 27 23:27:43 fedora sudo[11558]:  backup1 : user NOT in sudoers ;
TTY=pts/1 ; PWD=/home/backup1 ; USER=root ; COMMAND=/usr/sbin/tar -xz
f /backup/swift/swift-lt-2026-03-27.tar.gz -C /
Mar 27 23:28:40 fedora sudo[11819]:   admin1 : TTY=pts/1 ; PWD=/home/
```

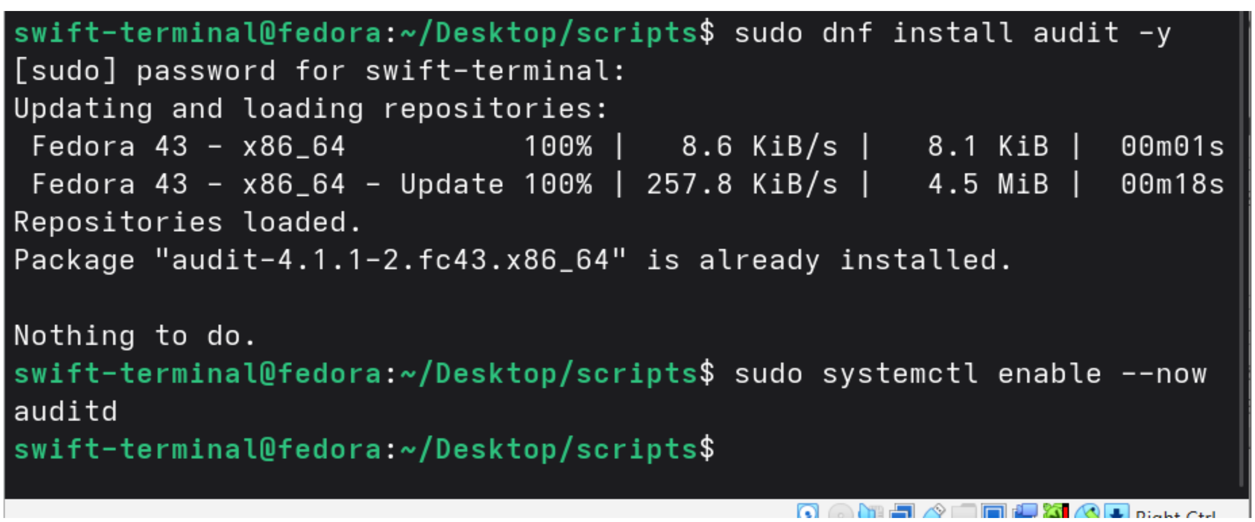
3. Setting up Auditing System

3.1 Critical Security Events

Event	Description	Target
Access to SWIFT files	Sensitive financial data access	/swift-lt
Configuration changes	System integrity risk	/etc
SSH login attempts	Unauthorized access	ssh
Privilege escalation	sudo/root usage	system
Backup modification	Data tampering	/backup
Database access	Metadata integrity	PostgreSQL

3.2 Install auditd

sudo dnf install audit -y
sudo systemctl enable --now auditd



3.3 Audit Rules

Edit:

sudo nano /etc/audit/rules.d/audit.rules

Add rules:

Monitor SWIFT files:

-w /swift-lt/ -p rwa -k swift_files

Monitor nginx config:

-w /etc/nginx/ -p wa -k nginx_changes

Monitor backend:

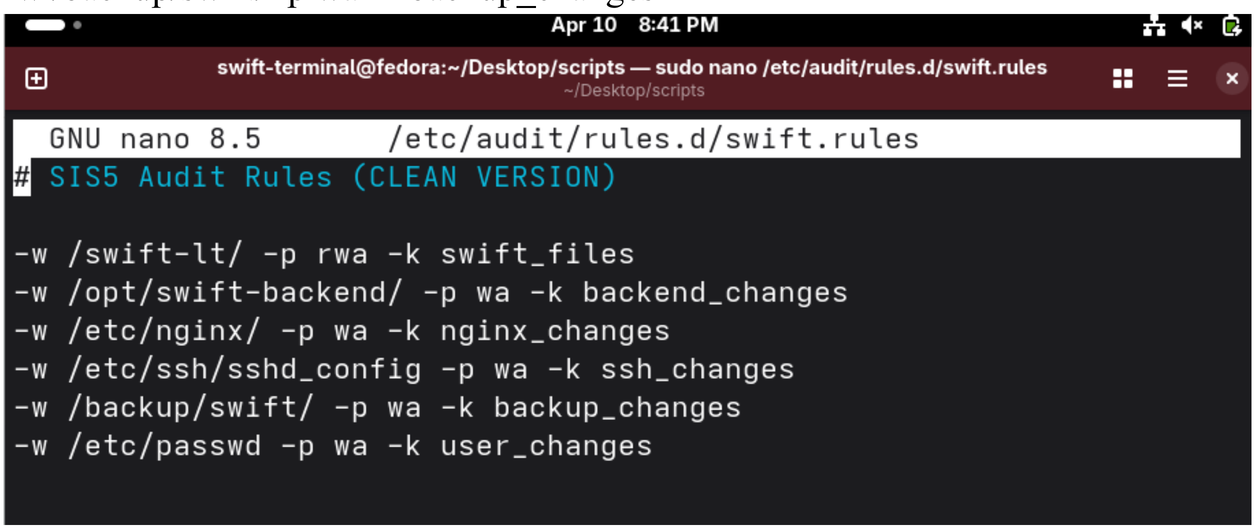
-w /opt/swift-backend/ -p wa -k backend_changes

Monitor SSH config:

-w /etc/ssh/sshd_config -p wa -k ssh_changes

Monitor backups:

-w /backup/swift/ -p wa -k backup_changes



Restart:

sudo systemctl restart auditd

3.4 Testing Audit System

Test 1 — Access SWIFT files

ls /swift-lt/LT1

Check:

ausearch -k swift_files

```
swift-terminal@fedora:~/Desktop/scripts$ sudo ls /swift-lt/LT1
test.txt  text.txt
swift-terminal@fedora:~/Desktop/scripts$ sudo ausearch -k swift_files
----
time->Fri Apr 10 19:42:59 2026
type=CONFIG_CHANGE msg=audit(1775832179.180:596): auid=4294967295 ses
=4294967295 subj=system_u:system_r:unconfined_service_t:s0 op=add_rul
e key="swift_files" list=4 res=1
----
time->Fri Apr 10 19:42:59 2026
type=CONFIG_CHANGE msg=audit(1775832179.180:597): auid=4294967295 ses
=4294967295 subj=system_u:system_r:unconfined_service_t:s0 op=add_rul
e key="swift_files" list=4 res=0
----
time->Fri Apr 10 20:35:18 2026
type=CONFIG_CHANGE msg=audit(1775835318.794:1513): auid=1000 ses=3 su
bj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 op=remove_ru
le key="swift_files" list=4 res=1
----
time->Fri Apr 10 20:37:28 2026
type=PROCTITLE msg=audit(1775835448.280:1573): proctitle=2F7362696E2F
617564697463746C002D52002F6574632F61756469742F61756469742E72756C6573
type=SYSCALL msg=audit(1775835448.280:1573): arch=c000003e syscall=44
 success=yes exit=1076 a0=3 a1=7ffd131ca2f0 a2=434 a3=0 items=0 ppid=
6091 pid=6110 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid
```

Test 2 — Modify nginx config

sudo nano /etc/nginx/nginx.conf

Check:

ausearch -k nginx_changes

```
swift-terminal@fedora:~/Desktop/scripts$ sudo nano /etc/nginx/nginx.c
onf
swift-terminal@fedora:~/Desktop/scripts$ ausearch -k nginx_changes
Error opening config file (Permission denied)
NOTE - using built-in end_of_event_timeout: 2
NOTE - using built-in logs: /var/log/audit/audit.log
Error opening /var/log/audit/audit.log (Permission denied)
swift-terminal@fedora:~/Desktop/scripts$ sudo ausearch -k nginx_chang
es
----
time->Fri Apr 10 20:37:28 2026
type=PROCTITLE msg=audit(1775835448.287:1575): proctitle=2F7362696E2F
617564697463746C002D52002F6574632F61756469742F61756469742E72756C6573
type=PATH msg=audit(1775835448.287:1575): item=0 name="/etc/nginx" in
ode=209239 dev=00:20 mode=040755 ouid=0 ogid=0 rdev=00:00 obj=system_
u:object_r:httpd_config_t:s0 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe
=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1775835448.287:1575): cwd="/home/swift-terminal/De
sktop/scripts"
type=SOCKADDR msg=audit(1775835448.287:1575): saddr=10000000000000000
0000000
type=SYSCALL msg=audit(1775835448.287:1575): arch=c000003e syscall=44
 success=yes exit=1080 a0=3 a1=7ffd131ca2f0 a2=438 a3=0 items=1 ppid=
6091 pid=6110 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid
=0 fsgid=0 tty=pts1 ses=3 comm="auditctl" exe="/usr/bin/auditctl" sub
j=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 key=(null)
type=CONFIG_CHANGE msg=audit(1775835448.287:1575): auid=1000 ses=3 su
bj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 op=add_rule
```

Test 3 — Failed login attempt

Try wrong SSH password.

Check:

ausearch -m USER_LOGIN

```
swift-terminal@fedora:~/Desktop/scripts$ sudo ausearch -m USER_LOGIN
----
time->Sat Feb  7 10:30:46 2026
type=USER_LOGIN msg=audit(1770442246.419:164): pid=1947 uid=0 auid=10
00 ses=2 subj=system_u:system_r:xdm_t:s0-s0:c0.c1023 msg='uid=1000 ex
e="/usr/libexec/gdm-session-worker" hostname=? addr=? terminal=? res=
success'
----
time->Sat Feb  7 15:14:55 2026
type=USER_LOGIN msg=audit(1770459295.752:158): pid=1700 uid=0 auid=10
00 ses=2 subj=system_u:system_r:xdm_t:s0-s0:c0.c1023 msg='uid=1000 ex
e="/usr/libexec/gdm-session-worker" hostname=? addr=? terminal=? res=
success'
----
time->Fri Feb 13 22:17:21 2026
type=USER_LOGIN msg=audit(1771003041.620:137): pid=1613 uid=0 auid=42
94967295 ses=4294967295 subj=system_u:system_r:xdm_t:s0-s0:c0.c1023 m
sg='uid=1000 exe="/usr/libexec/gdm-session-worker" hostname=? addr=?
terminal=? res=failed'
----
time->Fri Feb 13 22:17:27 2026
type=USER_LOGIN msg=audit(1771003047.753:143): pid=1682 uid=0 auid=42
94967295 ses=4294967295 subj=system_u:system_r:xdm_t:s0-s0:c0.c1023 m
sg='uid=1000 exe="/usr/libexec/gdm-session-worker" hostname=? addr=?
terminal=? res=failed'
```

Test 4 — Backup modification

sudo touch /backup/swift/test.txt

Check:

ausearch -k backup_changes

3.5 Observability Integration (Important)

From SIS1–SIS4, your system already includes:

- rsyslog → centralized logs
- Grafana + node_exporter → metrics monitoring
- journald → system logs

This creates **full observability stack**:

Layer	Tool
Logs	rsyslog + journald
Metrics	node_exporter
Visualization	Grafana
Security audit	auditd

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4. Security Advantages

- Logs stored on separate machine → attacker cannot erase evidence
- Full traceability of user and admin actions
- Detection of:
  - brute-force attacks
  - unauthorized file access
  - config tampering
- Compliance with financial security requirements

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5. Conclusions

In this study, a complete monitoring and observability system was successfully implemented for the Secure SWIFT Logical Terminal Infrastructure.

A centralized logging system was configured using rsyslog, ensuring that all system and application logs from the swift-terminal server are securely forwarded to the swift-support server. This design improves security by separating log storage from the main system.

Several log analysis scripts were created to monitor critical system events such as failed login attempts, web server errors, and access to sensitive SWIFT storage directories.

Additionally, a Linux auditing system was implemented using auditd. Critical security events were identified, audit rules were defined, and multiple test scenarios confirmed that all important actions are properly recorded and traceable.

Combined with the monitoring tools configured in SIS3 (Grafana and Node Exporter), the system now provides full observability, enabling real-time monitoring, security auditing, and incident investigation.

This completes the final layer of the defense-in-depth architecture defined in SIS1, ensuring that the system is not only secure but also fully observable and auditable.

Layer	**Tool / System**	**Runs on Machine**	**What it monitors**	**How to check**	**Example command**
📜 Logging (centralized)	rsyslog (client)	swift-terminal	Sends all system/service logs	Check forwarding config	`cat /etc/rsyslog.conf
📜 Logging receiver	rsyslog server	swift-support	Collects logs from all machines	Check log files	ls /var/log/remote/
📜 System logs	journald	swift-terminal	Kernel + system services	View logs	journalctl -xe
📜 SSH logs	system logs	swift-terminal	Login attempts (security)	Check auth logs	`sudo cat /var/log/secure
📜 Nginx logs	nginx	swift-terminal	Web access + errors	Check logs	sudo tail -f /var/log/nginx/access.log
📜 Backend logs	systemd journal	swift-terminal	FastAPI service behavior	View service logs	journalctl -u swift-backend
📜 Audit system	auditd	swift-terminal	File access + security events	Search audit logs	sudo ausearch -k swift_files
📜 Audit rules	auditctl / augenrules	swift-terminal	Defines monitored targets	List rules	sudo auditctl -l
📊 Metrics system	node_exporter	swift-support	CPU, RAM, disk, system stats	Check metrics	curl http://localhost:9100/metrics

📊 Monitoring dashboard	Grafana	swift-support	Visualization of metrics	Open web UI	http://<support-ip>:3000
💾 Backup monitoring	cron + rsync script	swift-support	Backup execution status	Check logs	cat /var/log/swift-backup.log
💾 Backup storage	filesystem	swift-support	Stores encrypted backups	List backups	ls /backup/swift/
🧪 Firewall monitoring	firewalld	both machines	Network access control	Check rules	sudo firewall-cmd --list-all

On swift-terminal (security + logs)

sudo auditctl -l
sudo ausearch -k swift_files
journalctl -u swift-backend
sudo tail /var/log/secure

On swift-support (monitoring + backup)

ls /var/log/remote/
cat /var/log/swift-backup.log
curl http://localhost:9100/metrics

Grafana check

http:// 10.156.104.194:3000

APP used:

swift-terminal (Data + Security Source)

- journald
- rsyslog (client)
- auditd

- auditctl / ausearch / augenrules
- nginx (logs source)
- backend logs (FastAPI via systemd)

swift-support (Monitoring + Centralization)

- rsyslog (server)
- Grafana
- node_exporter